

NMOS TRANSISTOR CHARACTERISTICS USING CADENCE VIRTUOSO

1. AIM

To design and simulate an NMOS transistor using **Cadence Virtuoso** and study its electrical characteristics by performing a DC sweep of the gate-to-source voltage while maintaining a constant drain-to-source voltage.

2. SOFTWARE AND TOOLS USED

The following tools and technology library were used for the simulation:

- **Cadence Virtuoso**
- **Cadence ADE L (Analog Design Environment L)**
- **Cadence ViVA (Visualization and Analysis)**
- **GPD90 technology library**

The schematic was designed in Cadence Virtuoso, the simulation was configured using ADE L, and the simulation waveforms were viewed using Cadence ViVA.

3. DESIGN SPECIFICATIONS

The NMOS transistor was configured using the GPD90 technology library with the following parameters:

Parameter	Value
Technology	GPD90
Total Width ()	1 μm
Finger Width	1 μm
Channel Length ()	100 nm
Number of Fingers ()	1
Multiplier ()	1
Drain-to-Source Voltage ()	0.6 V
Gate-to-Source Voltage ()	0–1.2 V

Since the number of fingers is one, the total transistor width is:

Therefore, the NMOS transistor used in the simulation has a width of **1 μm** and a channel length of **100 nm**.

4. CIRCUIT DESCRIPTION

The NMOS transistor was configured with two DC voltage sources to provide the required gate-to-source and drain-to-source voltages.

The **gate terminal** of the NMOS is connected to the voltage source. The **drain terminal** is connected to the voltage source. The **source terminal** is connected to ground, and the **body/bulk terminal** is also connected to ground.

The voltage sources were taken from the analogLib library and were assigned the design variables vgs and vds.

The circuit configuration allows the gate-to-source voltage to be varied while maintaining a constant drain-to-source voltage.

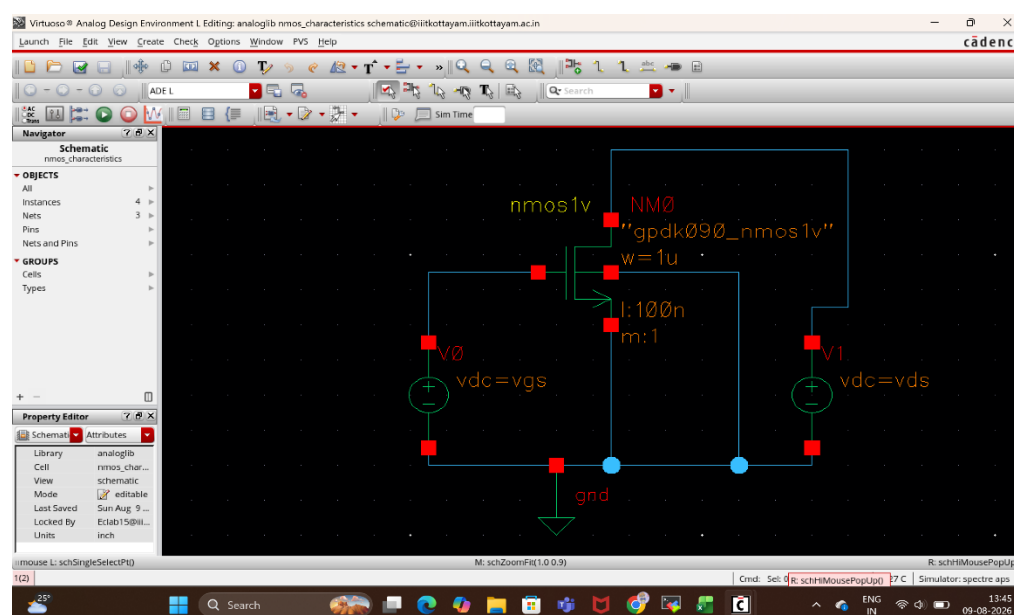


Figure 1: NMOS transistor test circuit designed in Cadence Virtuoso.

5. DESIGN VARIABLE CONFIGURATION

The DC voltage sources were configured using design variables so that their values could be controlled directly from Cadence ADE L.

The following design variables were used:

Vgs	0V
Vds	0.6V

For the initial simulation setup, the variables were assigned:

The vgs variable was subsequently selected as the sweep variable for the DC analysis, while vds was maintained at a constant value of 0.6 V.

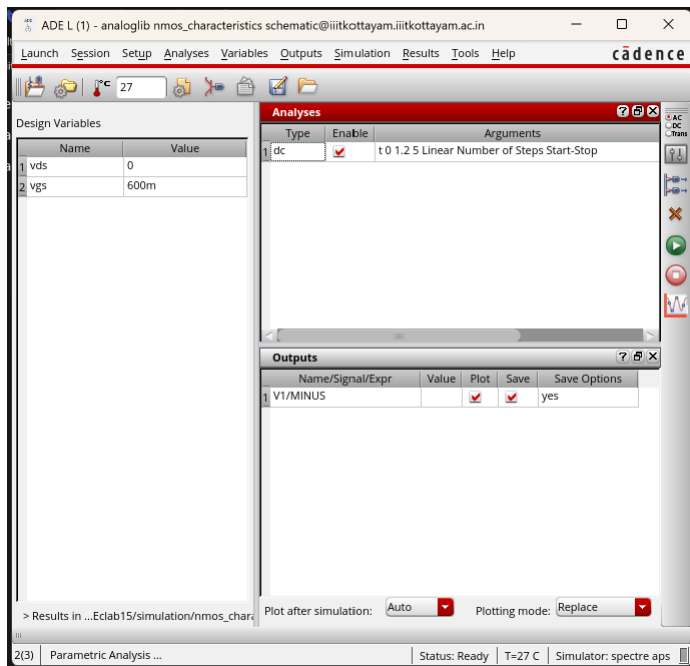


Figure 2: Design variables configured in Cadence ADE L.

7. SIMULATION EXECUTION

After configuring the DC analysis, the simulation was executed from Cadence ADE L.

The simulator performed the DC sweep by incrementally varying from 0 V to 1.2 V while keeping fixed at 0.6 V.

The resulting simulation data was transferred to the Cadence ViVA environment for waveform visualization and analysis.

8. SIMULATION RESULTS

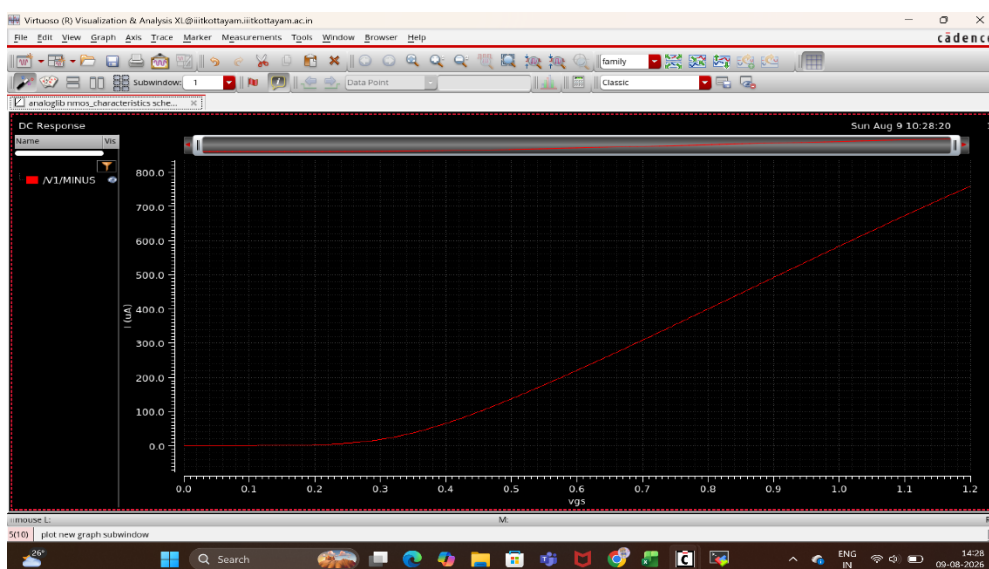


Figure 3: ID vs VGS



Figure 4: ID vs VDS

9. OBSERVATIONS

The following observations were made from the simulation:

1. The NMOS transistor was successfully configured using the GPD90 technology library.
2. The transistor was designed with a total width of 1 μm and a channel length of 100 nm.
3. The source and body terminals were connected to ground.
4. The gate and drain voltages were controlled using DC voltage sources.
5. The voltage sources were successfully controlled using the design variables vgs and vds.
6. A DC sweep was configured with vgs as the sweep variable.
7. was varied from 0 V to 1.2 V.
8. was maintained at a constant value of 0.6 V.
9. The DC simulation completed successfully in Cadence ADE L.
10. The resulting simulation waveforms were visualized using Cadence ViVa.

10. CONCLUSION

The NMOS transistor was successfully designed and simulated using **Cadence Virtuoso with the GPD90 technology library**.

Github link : https://github.com/mohanapriyams/AnalogElectronicsLab_2025BEC0041.git